MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 10545704

(ACQUISITION TIME DELAY)

1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of specification MIL-C-50785 comprise the complete requirements for the Acquisition Time Delay Circuit Card Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitatim for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MiL-C-S0785	Circuit Card Assemblies, General Specification for
MIL-c-50785/l	Circuit Card Assembly, Voltage Comparator

DRAWINGS

U. S. Army, Frankford Arsenal

10545425	Sight Current Generator
10545455	Receptacle
10545481	Circuit Card Assembly, Voltage Comparator
10545704	Circuit Card Assembly, Acquisition Time Delay

PACKAGING DATA SHEET

10545704	Packaging of Circuit Card Assembly, Acquisition Time Delay

(Copies of specification, standards, drawings, and publications required by Suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 128S

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3. REQUIREMENTS

3.1 Item definition. This assembly is a single ended printed circuit board consisting of a ramp generator, an operational amplifier with an input clamp, switching network and a voltage divider. the assembly is used to generate the acquisition time delay signal when installed in the **Sight Current Generator** Drawing Number 10545425.

3.1.1 Major component list. The components of this assembly shall be as specified on Engineering Parts List (EPL) 10545704.

3.1.2 Government furnished property list. Not applicable

3.2 **Characteristics**

3.2.1 *Perfomance*. *The* Assembly **shall, when supplied the specified inputs, be** capable of performing function a, b, c and d.

a. Generate the linear voltage ramp input required to develop the Acquisition Time Delay by means of the Al Ramp Generator.

b. **Amplify and clamp the range voltage input required to** develop the acquisition time delay by means of the **A2** amplifier.

c. Control the ramp generator and the acquisition time delay outputs through the switching network.

d. Provide two reference voltages for the **"READY TO FIRE"** light circuitry in the sight current generator by means of a divider.

3.2.1.1 Input and Output levels

3.2.1.1.1 Input power. Input power shall be supplied to the assembly as specified in Table 1.

Table 1. Input Power

Pin Number	Voltage
4 and 25	+15 vdc ± 2.0 percent
14	-15 vdc ± 2.0 percent
24	+24 vdc ± 2.0 percent
I and 28	Ground

3.2.1.1.2 Output levels. The requirements in **3.2.1.1.2.1 through 3.2.1.1.2.9** shall be met w en supplied with the input voltage (Vin] specified in Table X, under the conditions of Figure 1.

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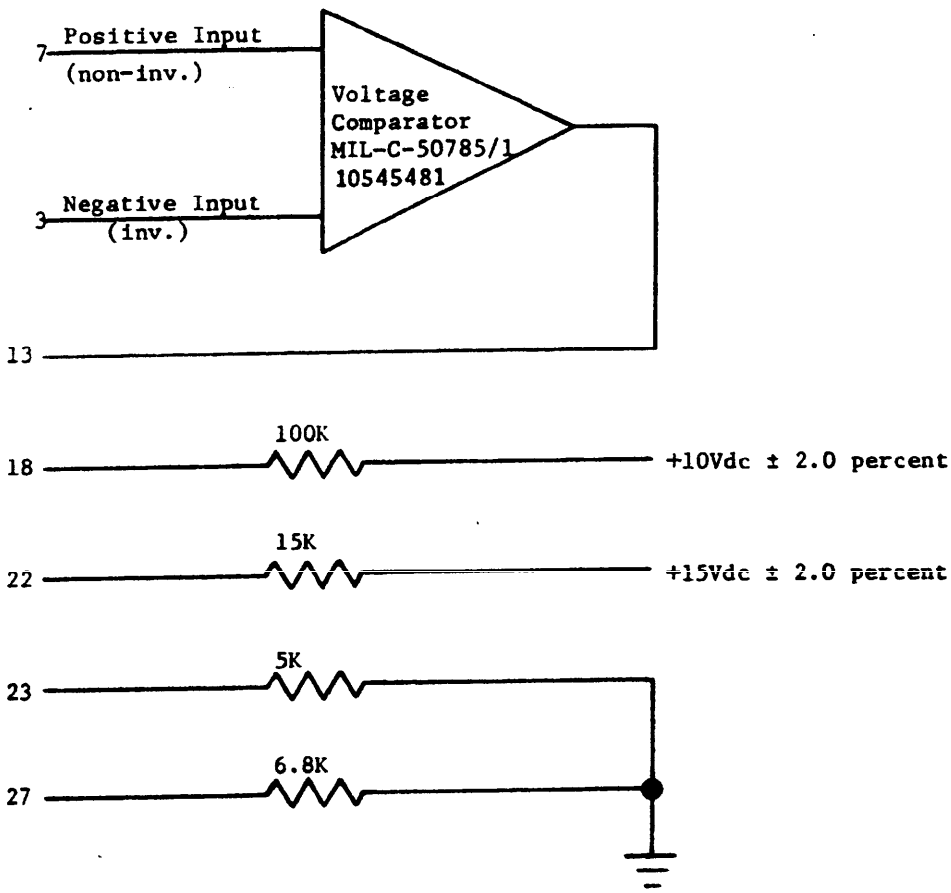


Figure 1. Acquisition Time Delay Board Interface

3.2.1.1.2.1 Pin 3 output. The output shall be [(+0.833 Vin)] ± 1.0 percent between pin 3 and pin one with a Vin ≤ 10 Vdc applied between pin 2 and pin one. The output shall be between 9 Vdc and 12 Vdc with a Vin between + 10 Vdc and +15 Vdc applied between pin 2 and pin one. The output shall be zero Vdc ± 40 millivolts with a zero Vin applied between pin 2 and pin one.

3.2.1.1.2.2 Pin 6 output. The ratio of the output between pin 6 and pin one to the input between pin 4 and pin one shall be 0.2008 ± 0.2 percent.

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3.2.1.1.2.3 Pin 7 output. The output between pin 7 and pin one shall be a linear ramp such that the requirements of 3.2.1.1.2.8 are met.

3.2.1.1.2.4 Pin 9 output. The ratio of the output between pin 9 and pin one to the input between pin 4 and pin one shall be 0.4980 ± 0.2 percent.

3.2.1.1.2.5 Pin 16 output. The output shall be zero Vdc ± 300 millivolts between pin 16 and pin 28 with pin eleven grounded.

3.2.1.1.2.6 Pin 18 output. The output shall be + 10 Vdc 2 10 millivolts between pin 18 and pin 28 with pin eleven grounded.

3.2.1.1.2.7 Pin 22 uutput. The output shall be +15 Vdc ± 5 percent between pin 22 and pin 28 with pin eleven grounded and zero Vdc ± 10 millivolts during the interval the output between pin 23 and pin 28 is + 15 Vdc.

3.2.1.1.2.8 Pin 23 output. The outputs between pin 23 and pin 28 shall be as specified in a and b.

a. -0.450 Vdc ± 150 millivolts with pin 18 grounded and with pin eleven grounded or open.

b. + 10 Vdc to +15 Vdc pulse initiated when in eleven is switched from ground to open, with a time duration (0.186 Vin -0.34s ± 0.25) second, where Vin is the voltage applied between pin 2 and one, for values of Vin ranging from 0.8 Vdc to 10 Vdc.

3.2.1.1.2.9 Pin 27 output. The output shall be zero Vdc ± 10 millivolts between pin 27 and pin 28 with pin eleven grounded, and +24 Vdc ± 2.0 percent during the interval the output between pin 23 and pin 28 is +15 Vdc.

3.2.1.2 Environmental. The assembly shall meet the environmental requirements of specification MIL-C-S0785.

3.2.2 Physical characteristic. The physical characteristics shall be as specified in drawing 10545704 and the engineering parts list.

3.3 Design and construction

3.3.1 Production drawings. This assembly shall be fabricated in accordance with the requirements of specification MIL-C-50785, drawing 10545704 and associated lists.

3.3.2 Standards of Manufacture. The standards of manufacture shall be those specified in specification MIL-C-50785.

3.3.3 Workmanship. The workmanship shall be as specified in specification MIL-C-50785.

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4. QU

4.1 Gene

4.1.1 Res The responsibility for inspection shall be as spec 50785.

4.1.2 Spe The assembly shall undergo first article and contrc 4.1.2 of specification MIL-C-50785.

4.2 Quali The assembly shall be inspected and tested in ication MIL-C-50785.

4.2.1 Tes

4.2.1.1 P supply shall be 15 ± 0.1 Vdc, 100 milli-amperes (man 10 millivolts (mv) rms or 100 mv peak-to-peak.

4.2.1.2 P supply shall be -15 ±0.1 Vdc, 100 ma or greater or 100 mv peak-to-peak.

4.2.1.3 P supply shall be 24 ± 0.5 Vdc, 100 ma or greater.

4.2.1.4 P supply shall be +10.0 ± 0.010 Vdc, 10 ma or grea

4.2.1.5 P supply shall be zero to + 15 Vdc with an accuracn to ± .01 percent, 100 ma or greater.

4.2.1.6 Dl voltmeter shall have a voltage accuracy and resolu from zero to 100 Vdc. Input impedance greater ths cales.

4.2.1.7 Ristors are required with ratings of 100,000 ohd 5000 ohms, ± 5.0 percent.

4.2.1.8 Se pole single throw are required.

4.2.1.9 Vit Card Assembly, Voltage Comparator MIL-C-5078 red.

4.2.1.10 be shall have voltage, time accuracy and resolu with a rise time of 100 nanoseconds or better Voltage sensitivity shall include the range ts/cm.

4.2.2 Fur

4.2.2.1 Supplies shall be connected to the assembly under t 2 using receptable 10545455. Power supply and ge comparator shall be as specified in MIL-C-5(

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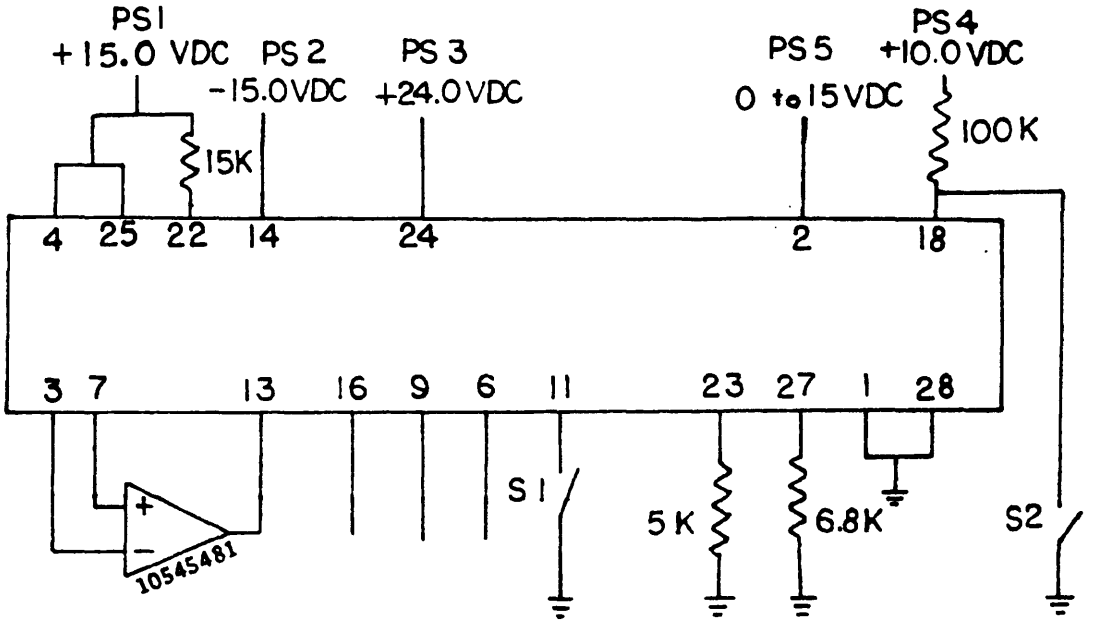


Figure 2. Test Connections

4.2.2.2 Test procedure. Compliance with the requirements of 3.2.1.1.2.1 through 3.2.1.1.2.9 shall be verified as follows:

a. Adjust power supply PS_5 as specified in Table II. Close switch S1 (pin 11 grounded). The voltage on pin 3 shall be as specified. The voltage on pin 23 as measured with the oscilloscope shall be -0.450 ± 0.15 Vdc, rising to 10 to 15 Vdc when switch S1 is opened, then returning to -0.450 ± 0.15 Vdc after the specified delay. The voltage on pin 27 shall rise from zero ± 0.010 Vdc to 24 ± 2 Vdc only when the pin 23 voltage rises to 10 to 15 Vdc. The voltage on pin 22 shall drop from 15.0 ± 0.5 Vdc to zero ± 0.010 Vdc only when the pin 23 voltage rises to 10 to 15 Vdc.

Table II. Input Adjustment

Pin 2 Input Vdc	Pin 3 Output Vdc	Pin 23 Time Delay in Seconds
Zero ± 0.010	Zero ± 0.040	No pulse
+5 ± 0.010	+ 4.167 ± 0.042	0.33 to 0.84
+7.5 ± 0.010	+6.250 ± 0.063	0.80 to 1.30
+10 ± 0.010	+8.333 ± 0.083	1.26 to 1.77
+15 ± 0.015	+9 to +12	--

b. Adjust PS5 to +10 ± 0.010 Vdc. Close switch S2 (pin 18 grounded). The voltage on pin 23 shall remain at -0.450 ± 0.150 Vdc with switch S1 open or closed.

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c* With switch S1 closed and switch S2 open the voltage on pin 16 shall be zero ± 0.30 Vdc and the voltage on pin 18 shall be 10.0 ± 0.010 Vdc.

d. The ratio of voltage on pin 6 to the voltage on pin 4 shall be as specified in 3.2.1.1.2.2.

e. The ratio of voltage on pin 9 to the voltage on pin 4 shall be as specified in 3.2.1.1.2.4.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging and marking shall be in accordance with the level A requirements of Packaging Data Sheet (PDS), AMC Form 1029, 10545704.

6. NOTES

6.1 The notes specified in the general specification MIL-C-50785 are applicable to this specification.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES | NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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